

Project Title: Field-Granular Schema-Annotated Trust for Offline-First Synchronization

1. Introduction & Problem Statement

Vulnerability in Current Systems

Offline-first distributed databases, such as CouchDB and PouchDB, are highly effective for mobile and IoT applications, but they carry a known security flaw. To resolve synchronization conflicts, these systems default to a Last-Write-Wins (LWW) strategy. This mechanism relies entirely on timestamps. As established in our baseline research[1], this is a critical architectural vulnerability. Any compromised node on the network can forge a future timestamp to ensure its malicious update is blindly prioritized and accepted as the authoritative "master" record.

- **Example of Vulnerability:** Consider an offline-first healthcare application where a patient's medication dosage is stored locally. A compromised mobile device modifies the critical dosage offline and intentionally forges a future timestamp. Upon reconnection, the default LWW protocol evaluates the conflict and unconditionally accepts the malicious update, which could cause severe physical harm.

The Granularity Mismatch

Previous attempts to fix this issue intercepted the sync process using global trust models. The foundational mathematics of these mechanisms rely on Weighted Majority Voting (WMV) to guarantee decision stability[2] and Beta trust frameworks for reputation calculation[3][4]. However, the baseline application of a uniform/single trust threshold (e.g., $\tau=0.6$) across an entire document is still a severe limitation in our previous research[1].

This lack of granularity causes two critical failures:

- **Over-trusting (Reputation Laundering):** A malicious actor builds a high global reputation by consistently providing honest updates to low-stakes administrative fields, such as updating UI themes or display names. Because the system applies a single document-level threshold, this inflated global authority allows the malicious node to bypass security and corrupt high-stakes data—like a patient's surgical history—within the same document.
 - **Under-trusting (System Rigidity):** To prevent reputation laundering, an administrator might set the global trust threshold conservatively high (e.g., $\tau=0.85$) to protect sensitive clinical data. Consequently, a newly registered, legitimate nurse with only a moderate baseline trust score is blocked from making routine, low-risk updates, such as changing a patient's room number or notification preferences, resulting in operational friction and data loss.
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2. Proposed Architecture

To tackle this, this major project will develop a Node.js middleware synchronization interceptor that implements a **field-granular, schema-annotated trust framework**. The components of this framework are defined as follows:

- **Field-Granular:** Instead of resolving conflicts for an entire document unit at once, the system decomposes the document and evaluates source modifications independently for every specific attribute using dot-notation pathing (e.g., medication.dosage).
- **Schema-Annotated:** The practice of embedding risk parameters, security policies, and required evidence thresholds directly into the structural metadata of the data model, extending standard JSON Schema.
- **Trust Framework:** The overarching algorithmic engine—combining Bayesian statistical history tracking and weighted majority voting—that assesses historical source reliability to resolve data conflicts, removing the reliance on arbitrary timestamps.

Core System Components & Justifications:

A. Schema-Annotated Metadata ("Compliance-as-Code")

We will extend standard JSON Schema to allow domain experts to directly embed risk parameters into the data model using custom keywords like $\tau_{\{f\}}$ (required cumulative trust threshold) and $\lambda_{\{f\}}$ (global-to-field reputation inheritance).

- *Justification:* This approach is directly informed by recent research on machine-readable compliance[6], which advocates for embedding structural risk and compliance rules directly into data models rather than hardcoding them into application logic. By extending the schema, domain experts can directly define the exact risk rules for every individual attribute. This ensures that "Golden Record" security policies are versioned alongside the data and consistently enforced across all synchronization points.

B. Hierarchical Bayesian Trust Engine

Instead of starting from zero for every new field, the system uses a hierarchical posterior model where field-level trust estimates can inherit from the user's global Beta distribution.

- *Justification:* Moving from document-level to field-level trust introduces a severe "cold-start" problem (data sparsity) when a user updates a specific field for the first time. To solve this, we adapt established mathematics for hierarchical Bayesian modeling[5]. This allows a specific field to intelligently inherit trust from the user's overall global reputation, scaled by the schema's $\lambda_{\{f\}}$ weight. High-risk fields ignore global trust ($\lambda_{\{f\}} \rightarrow 0$) to isolate risk, while low-risk administrative fields fully leverage it ($\lambda_{\{f\}} = 1$) to maintain high system availability.

C. Per-Field Weighted Majority Voting (WMV)

- *Objectives:* To optimize decision-making accuracy independently for each individual data

attribute during a synchronization conflict.

- *How it works:* Drawing on methodologies for structural risk discovery[7], the middleware interceptor isolates structural risk by decomposing the incoming document using dot-notation (e.g., user.preferences.theme). For each isolated field, it aggregates claims and executes log-odds weighted voting based on the individual sources' calculated field-trust scores:

$$w_{u,f} \propto \log \left(\frac{\hat{p}_{u,f}}{1 - \hat{p}_{u,f}} \right)$$

The winning claim is committed to the canonical master database only if its cumulative weight surpasses the specific τ_f threshold defined in the schema registry.

- *Why it attains the objective:* This mechanism is mathematically supported by *Stability of Correctness* and *Stability of Optimality* bounds[2], which guarantee that decision accuracy remains highly resilient even when WMV elections are run using dynamically estimated historical trust data.

3. System Architecture & Key Outcomes

Significance of Partial Commits

The most significant operational feature of this architecture is the "partial commit". In traditional LWW or uniform WMV models, conflict resolution is an "all-or-nothing" process; a single unverified or suspicious field forces the system to reject the entire document revision. Partial commits empower the merge assembler to surgically accept highly confident, legitimate updates to safe fields (e.g., a display name) while rejecting untrusted modifications to sensitive fields (e.g., medication dosage) within the exact same sync cycle. This fundamentally maximizes system availability and prevents arbitrary data loss.

Evaluation Methodology & Target Metrics

Because evaluating offline-first conflict resolution requires a distributed environment, a core component of this project will be developing a custom Node.js simulation framework to empirically benchmark our proposed architecture.

We will conduct a comparative ablation study by simulating a network of edge nodes performing concurrent offline edits. To accurately test the system's resilience, we will inject adversarial nodes specifically programmed to execute "Reputation Laundering" attacks (building trust on low-stakes fields to subsequently attack high-stakes fields).

We will benchmark our proposed Field-Granular WMV against two control architectures:

1. **Document-Level LWW:** The standard default protocol (e.g., native PouchDB).
2. **Uniform WMV ($\tau=0.6$):** The document-level trust baseline established in our prior minor project research[1].

Target Objectives & Efficiency Proofs:

Based on the mathematical bounds of the Beta distribution and the stability guarantees of

WMV, our experimental design aims to prove the following target metrics:

- **Malicious Rejection Rate (Security):** Achieve a >99% rejection rate for malicious updates on critical risk tiers (compared to 0% in standard LWW and 97% in Uniform WMV).
- **False Positive Rate (Usability):** Reduce the FPR for legitimate, low-stakes administrative updates to <1% (solving the 3-5% "under-trusting" system rigidity of Uniform WMV).
- **Sync Latency (Efficiency):** Prove that decomposing JSON paths and calculating multi-level hierarchical weights adds nominal overhead. Because updating a Bayesian posterior and executing log-odds weighted calculations are constant-time $\mathcal{O}(1)$ operations, we target a total processing latency of 65-80ms per sync event.
- **Memory Overhead:** Empirically validate that trust tracking memory is mathematically bounded. Because tracking a Beta distribution only requires storing two 32-bit integers (α for successes, β for failures), we target a global memory footprint of strictly <200 bytes per user to ensure viability for IoT and mobile environments.

4. Core References

[1] *Trust-Aware Offline Synchronization System with Weighted Majority Voting* (Minor Project Report)

[2] Bai, S., et al. (2024). *Stability of Weighted Majority Voting under Estimated Weights*. Proceedings of AAMAS 2024.

[3] Jøsang, A., & Ismail, R. (2002). *The beta reputation system*. Proceedings of the 15th Bled Electronic Commerce Conference.

[4] Alsolami, Y., et al. (2022). *Lightweight trust management for iot*. Sensors.

[5] *An Efficient and Versatile Approach to Trust and Reputation using Hierarchical Bayesian Modelling*. ePrints Soton, University of Southampton, <https://eprints.soton.ac.uk/342467/1/habit.pdf>

[6] *Making AI Compliance Evidence Machine-Readable*. arXiv:2604.13767v1.

[7] *Search-Based Risk Feature Discovery in Document Structure Spaces under a Constrained Budget*. arXiv:2601.21608v1.